ON-BOARD DIAGNOSTIC SYSTEM PID/DATA MONITOR FUNCTION [SKYACTIV-D 1.5]

id0102q2142100

- The PID/DATA monitor items are shown below.

PID/DATA monitor item table

—: Not applicable

Item (definition)	Definition	Unit/Condition
AC_PRES	Refrigerant pressure	kPa, Bar, psi
AC_REQ	A/C request signal	Off/On
ACCS	A/C relay	Off/On
ALTT V*4	Generator output voltage	V
AMB_TEMP	Ambient air temperature	°C, °F
APP1	APP sensor No.1 voltage	V
	APP sensor No.1	%
APP2	APP sensor No.2 voltage	V
	APP sensor No.2	%
ARPMDES	Target engine speed	RPM
BARO	Barometric pressure	kPa, Bar, psi
BATT_CUR	Battery current	A
BATT_DAY	Number of days elapsed since current sensor initialization	—
BATT_RES	Battery internal resistance (estimated)	—
BATT_SOC	Battery charge condition (estimated)	%
BATT_TEMP	Battery temperature	°C, °F
BATT_V	Battery voltage	V
BBP	Power brake unit vacuum sensor	kPa, Bar, psi
BFP	Brake fluid pressure	kPa, Bar, psi
BOO	Brake switch (No.1 signal)	High/Low
BPA	Brake switch (No.2 signal)	High/Low
CACT11	Boost air temperature	°C, °F
CCV_ACT	Coolant control valve actual value	°(deg)
CCV_DSD	Coolant control valve desired value	°(deg)
CC_DIFP_WOA	Actual difference in phase between camshaft and crankshaft (no correction)	°(deg)
CLR_DIST	Distance after DTC cleared	—
CLU_CUT_SW*1	Starter interlock switch	Off/On
CLU_SW*1	CPP switch	Off/On
CPP*1	Clutch pedal position	%
CPP/PNP*1	Shift lever position	Off/On
DPF_LMP	Diesel particulate filter indicator light	Off/On
DPF_LMP_CNT	Number of times diesel particulate filter indicator light illuminates	—
DPF_REG_CNT	Diesel particulate filter regeneration count	—
ECT	Engine coolant temperature	°C, °F
EOL*4	Engine oil level	mm
EOT2*4	Engine oil temperature from engine oil level sensor	°C, °F
EOP	Engine oil pressure	kPa, Bar, psi
EOT	Engine oil temperature	°C, °F
E_WP	Electric water pump desired value	%
EXHPRES1	Exhaust gas pressure (No.1)	kPa, Bar, psi
EXHPRESS_DIF	Exhaust gas pressure (No.2)	kPa, Bar, psi
EXHP_DIF_AVE	Exhaust gas pressure (No.2) average	kPa, Bar, psi
EXHSHUT_ACT	Exhaust shutter valve actual value	%
EXHSUT_DSD	Exhaust shutter valve desired value	%
EXHTEMP1	Exhaust gas temperature (No.1)	°C, °F
EXHTEMP2	Exhaust gas temperature (No.2)	°C, °F
EXHTEMP3	Exhaust gas temperature (No.3)	°C, °F
EXHTEMP4	Exhaust gas temperature (No.4)	°C, °F
FAN_DUTY	Fan control module	%
FI_LRN_11	Fuel injection learning value (fuel injector No.1 at 35 MPa {357 kgf/cm^2, 5076 psi})	—(µs)
FI_LRN_12	Fuel injection learning value (fuel injector No.2 at 35 MPa {357 kgf/cm^2, 5076 psi})	—(µs)
FI_LRN_13	Fuel injection learning value (fuel injector No.3 at 35 MPa {357 kgf/cm^2, 5076 psi})	—(µs)
FI_LRN_14	Fuel injection learning value (fuel injector No.4 at 35 MPa {357 kgf/cm^2, 5076 psi})	—(µs)

Item (definition)	Definition	Unit/Condition
FI_LRN_21	Fuel injection learning value (fuel injector No.1 at 65 MPa {663 kgf/cm^2, 9427 psi})	—(μs)
FI_LRN_22	Fuel injection learning value (fuel injector No.2 at 65 MPa {663 kgf/cm^2, 9427 psi})	—(μs)
FI_LRN_23	Fuel injection learning value (fuel injector No.3 at 65 MPa {663 kgf/cm^2, 9427 psi})	—(μs)
FI_LRN_24	Fuel injection learning value (fuel injector No.4 at 65 MPa {663 kgf/cm^2, 9427 psi})	—(μs)
FI_LRN_31	Fuel injection learning value (fuel injector No.1 at 95 MPa {969 kgf/cm^2, 13779 psi})	—(μs)
FI_LRN_32	Fuel injection learning value (fuel injector No.2 at 95 MPa {969 kgf/cm^2, 13779 psi})	—(μs)
FI_LRN_33	Fuel injection learning value (fuel injector No.3 at 95 MPa {969 kgf/cm^2, 13779 psi})	—(μs)
FI_LRN_34	Fuel injection learning value (fuel injector No.4 at 95 MPa {969 kgf/cm^2, 13779 psi})	—(μs)
FI_LRN_41	Fuel injection learning value (fuel injector No.1 at 125 MPa {1275 kgf/cm^2, 18130 psi})	—(μs)
FI_LRN_42	Fuel injection learning value (fuel injector No.2 at 125 MPa {1275 kgf/cm^2, 18130 psi})	—(μs)
FI_LRN_43	Fuel injection learning value (fuel injector No.3 at 125 MPa {1275 kgf/cm^2, 18130 psi})	—(μs)
FI_LRN_44	Fuel injection learning value (fuel injector No.4 at 125 MPa {1275 kgf/cm^2, 18130 psi})	—(μs)
FI_LRN_51	Fuel injection learning value (fuel injector No.1 at 160 MPa {1632 kgf/cm^2, 23206 psi})	—(μs)
FI_LRN_52	Fuel injection learning value (fuel injector No.2 at 160 MPa {1632 kgf/cm^2, 23206 psi})	—(μs)
FI_LRN_53	Fuel injection learning value (fuel injector No.3 at 160 MPa {1632 kgf/cm^2, 23206 psi})	—(μs)
FI_LRN_54	Fuel injection learning value (fuel injector No.4 at 160 MPa {1632 kgf/cm^2, 23206 psi})	—(μs)
FIA_DSD	Supply pump flow desired value	—(mm^3/Stroke)
FIP_FL	Supply pump flow control current	A
FIP_SCV	Suction control valve	A
FP_DUTY	Supply pump duty cycle	%
FP_RCV	Fuel pressure relief valve operation time	—(ms)
FRP	Common rail pressure	kPa, Bar, psi
FRP_DSD	Common rail pressure desired value	kPa, Bar, psi
FRT	Fuel temperature inside the fuel supply line	°C, °F
GENVDS	Generator voltage desired	V
GLOW_P_V	Glow plug voltage	V
GP_LMP	Glow indicator light	On/Off
HTR11	A/F sensor heater control	%
HPEGR_ACT	HP-EGR control valve actual value	%
HPEGR_DSD	HP-EGR control valve desired value	%
IAT	Intake air temperature (No.1)	°C, °F
IAT13	Intake air temperature (No.2)	°C, °F
INGEAR*2	Gears are engaged	Off/On
INJ_AL_DIS	Distance travelled when automatic fuel injection amount learning	km, ft, mi
INJ_AL_FRQ	Number of times automatic fuel injection amount learning is completed	—
INJ_WL_DIS	Distance travelled when fuel injection amount learning at service factory	km, ft, mi
INJ_WL_FRQ	Number of times fuel injection amount learning is completed at service factory	—
INJ1_CMP	Fuel injector No.1 correction value	—(mm^3/Stroke)
INJ2_CMP	Fuel injector No.2 correction value	—(mm^3/Stroke)
INJ3_CMP	Fuel injector No.3 correction value	—(mm^3/Stroke)
INJ4_CMP	Fuel injector No.4 correction value	—(mm^3/Stroke)
I-Stop_OFF	i-stop OFF mode	Off/On
I-Stop_TRD	i-stop transmission D position selected status	Off/On
I-Stop_VSP	i-stop vehicle speed history flag	Off/On
I-Stop_VST	i-stop vehicle stop flag	Off/On
ISV_ACT	Intake shutter valve control actual value	°(deg)
ISV_DSD	Intake shutter valve control desired value	°(deg) %

Item (definition)	Definition	Unit/Condition
ISV_LRN_C	Intake shutter valve learning value (closed)	°(deg)
ISV_POS	Intake shutter valve	%
LPEGR_ACT	LP-EGR control valve actual value	%
LPEGR_DSD	LP-EGR control valve desired value	%
LOAD	Engine load	%
M_GEAR*1	Manual gear position	Neutral/1st gear/ 2nd gear/3rd gear/ 4th gear/5th gear/ 6th gear/Reverse/ Undefined/Auto/ In_Progress/YSF/ Error
MAF	Mass air flow	g/Sec
MAP	Manifold absolute pressure (No.2)	kPa, Bar, psi
MAP_DSD	Manifold absolute pressure (No.2) desired value	kPa, Bar, psi
MIL	Check engine light	Off/On
MIL_DIS	Travelled distance since check engine light illuminated	km, ft, mi
O2	Oxygen concentration in exhaust gas	%
O2S_IMP	A/F sensor element impedance	Ω
O2S11	A/F sensor	μA
		V
O2S11_CAL	A/F sensor calibration value	—
O2S11_MODE	A/F sensor activation status	—
OIL_DIL	Engine oil dilution amount	kg, lb
OIL_P_DUTY	Engine oil pressure control circuit duty cycle	%
OILCHG_DIS	Distance from the last engine oil change	km, ft, mi
PCVHC	Blow-by heater control	%
PM_ACC	PM accumulation amount	—(g/l, lb/ft^3)
PM_ACC_DSD	PM accumulation amount desired	—(g/l, lb/ft^3)
PM_GEN	PM generation amount	—(g/l, lb/ft^3)
PN_SW	Park/Neutral position determination	Open/Closed
PRE_CHG_DIS*3	Distance at precharge	km, ft, mi
REG_DIS	Distance since last diesel particulate filter regeneration	km, ft, mi
REV_SW	Reverse position determination	Off/On
RPM	Engine speed	RPM
VGTURB_RPM	Turbocharger with variable turbine geometry turbine speed	RPM
VGTURB_ACT	Turbocharger with variable turbine geometry position actual value	%
VGTURB_DSD	Turbocharger with variable turbine geometry position desired value	%
VPWR	Battery positive voltage	V
VSS	Vehicle speed	KPH, MPH

*1 : MTX
*2 : ATX
*3 : Without i-ELOOP
*4 : European (L.H.D., U.K.) specs.